

The Ordered Conditional Approximation and the Vecchia Approximation

Sam Power

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Introduction

The Vecchia approximation has emerged as a cornerstone of large-scale spatial modeling, particularly within the Gaussian process framework, by exploiting the sparsity of ordered conditional distributions to bypass the cubic complexity of exact joint inference. However, extending the Vecchia logic to general Markov Random Fields (MRFs) presents a significant challenge: while Gaussian conditionals are available in closed form, the conditional laws of a Gibbs measure are typically coupled through a global partition function, rendering them analytically intractable. The Ordered Conditional Approximation (OCA) addresses this limitation by localizing the system's Hamiltonian, effectively synthesizing tractable local models that account for both a sparsified history and the implicit influence of "future" observations. These notes provide a formal bridge between these two methods, examining the conditions under which local Hamiltonian decompositions yield valid approximations and outlining the step-by-step aggregation of local models into a global joint law.

Fundamentals of the Vecchia Approximation

In what follows, we consider a spatial random field $\{x_i : i \in \mathcal{V}\}$, where $\mathcal{V}$ is some index set accommodating a partial ordering $\prec$. Assume access also to a mapping λ which receives as input an index $i \in \mathcal{V}$, and outputs a subset of indices $\lambda(i) \subseteq \{j \in \mathcal{V} : j \prec i\}$, typically of much smaller size. One can think of these as the ‘most relevant’ ancestors of the vertex i . Given such a mapping, the basic Vecchia approximation is that

$$\mathbf{P}(dx) = \prod_{i \in \mathcal{V}} \mathbf{P}(dx_i \mid x_{\prec i}) \approx \prod_{i \in \mathcal{V}} \mathbf{P}(dx_i \mid x_{\lambda(i)}).$$

Let us take a moment to examine the quality of this approximation. Write out the autoregressive conditional distribution $\mathbf{P}(dx_i \mid x_{\prec i})$ by partitioning $\{j \in \mathcal{V} : j \prec i\} = \lambda(i) \sqcup \bar{\lambda}(i)$ to see that

$$\begin{aligned} \mathbf{P}(dx_i \mid x_{\prec i}) &= \mathbf{P}(dx_i \mid x_{\lambda(i)}, x_{\bar{\lambda}(i)}) \\ &= \frac{\mathbf{P}(dx_i, x_{\bar{\lambda}(i)} \mid x_{\lambda(i)})}{\int_{\mathcal{X}_i} \mathbf{P}(d\bar{x}_i, x_{\bar{\lambda}(i)} \mid x_{\lambda(i)})} \\ &= \frac{\mathbf{P}(dx_i \mid x_{\lambda(i)}) \cdot p(x_{\bar{\lambda}(i)} \mid x_i, x_{\lambda(i)})}{\int_{\mathcal{X}_i} \mathbf{P}(d\bar{x}_i \mid x_{\lambda(i)}) \cdot p(x_{\bar{\lambda}(i)} \mid \bar{x}_i, x_{\lambda(i)})} \\ &= \mathbf{P}(dx_i \mid x_{\lambda(i)}) \cdot \frac{p(x_{\bar{\lambda}(i)} \mid x_i, x_{\lambda(i)})}{\mathbf{E}\left[p(x_{\bar{\lambda}(i)} \mid X_i, x_{\lambda(i)}) \mid X_{\lambda(i)} = x_{\lambda(i)}\right]}, \end{aligned}$$

i.e.

$$\frac{d\mathbf{P}_i(\cdot \mid x_{\prec i})}{d\mathbf{P}_i(\cdot \mid x_{\lambda(i)})} = \frac{p(x_{\bar{\lambda}(i)} \mid x_i, x_{\lambda(i)})}{\mathbf{E}\left[p(x_{\bar{\lambda}(i)} \mid X_i, x_{\lambda(i)}) \mid X_{\lambda(i)} = x_{\lambda(i)}\right]}.$$

Some Error Analysis

Using this perspective, the approximation can be seen to be of reasonable quality, provided that under the distribution of $[X_i \mid X_{\lambda(i)} = x_{\lambda(i)}]$, it holds for all x that

$$p(x_{\bar{\lambda}(i)} \mid x_i, x_{\lambda(i)}) \approx \mathbf{E} \left[p(x_{\bar{\lambda}(i)} \mid X_i, x_{\lambda(i)}) \mid X_{\lambda(i)} = x_{\lambda(i)} \right],$$

i.e. provided that the sensitivity of the distribution of $X_{\bar{\lambda}(i)} \mid X_i, X_{\lambda(i)}$ to X_i is small. For example, under the (strong) assumption that

$$\forall x_i, x'_i, \quad \varepsilon \leq \frac{d\mathbf{P}(\cdot \mid x_i, x_{\lambda(i)})}{d\mathbf{P}(\cdot \mid x'_i, x_{\lambda(i)})} \leq \varepsilon^{-1},$$

it follows that

$$\varepsilon \leq \frac{d\mathbf{P}_i(\cdot \mid x_{\prec i})}{d\mathbf{P}_i(\cdot \mid x_{\lambda(i)})} \leq \varepsilon^{-1}$$

also. This opens the door to some quantitative bounds on the approximation quality of the Vecchia approximation.

Vecchia with Intractable Conditional Distributions

For Gaussian processes, the conditional distribution $\mathbf{P}(dx_i \mid x_{\lambda(i)})$ is available in closed form. For general Markov random fields, this is not usually the case, due to the presence of substantial intractability: assuming knowledge only of the unnormalised joint distribution, Bayes' theorem expresses $\mathbf{P}(dx_i \mid x_{\lambda(i)})$ as a ratio of two intractable integrals, viz. (being somewhat informal about measures and densities)

$$p(x_i \mid x_{\lambda(i)}) = \frac{p(x_i, x_{\lambda(i)})}{p(x_{\lambda(i)})} = \frac{\int_{\mathcal{X}_{(i \sqcup \lambda(i))}^{\mathfrak{C}}} p(x_i, x_{\lambda(i)}, \bar{x}_{(i \sqcup \lambda(i))}^{\mathfrak{C}}) d\bar{x}_{(i \sqcup \lambda(i))}^{\mathfrak{C}}}{\int_{\mathcal{X}_{\lambda(i)}^{\mathfrak{C}}} p(x_i, x_{\lambda(i)}, \bar{x}_{(i \sqcup \lambda(i))}^{\mathfrak{C}}) d\bar{x}_{\lambda(i)}^{\mathfrak{C}}}.$$

The Ordered Conditional Approximation (OCA) of Chakraborty-Katzfuss-Guinness proposes to resolve this difficulty by adding another layer of structured approximation.

The approach is as follows: assume now access to a mapping ρ which receives as input an index $i \in \mathcal{V}$, and outputs a subset of indices $\rho(i) \subseteq \{j \in \mathcal{V} : j \succeq i\}$, typically of small size. Note that here, the subset of indices looks into the future, rather than into the past. These indices should then be viewed as relevant descendant nodes of the vertex i .

Figure 1: A Sketch of the Neighbourhoods used in the OCA



Now, develop first the equality

$$\begin{aligned} \mathbf{P}(dx_i \mid x_{\lambda(i)}) &= \int_{\mathcal{X}_{\rho(i)}} \mathbf{P}(dx_i, d\bar{x}_{\rho(i)} \mid x_{\lambda(i)}) \\ &= \frac{\int_{\mathcal{X}_{\rho(i)}} \mathbf{P}(dx_i, d\bar{x}_{\rho(i)}, x_{\lambda(i)})}{\int_{\mathcal{X}_{i \sqcup \rho(i)}} \mathbf{P}(d\bar{x}_i, d\bar{x}_{\rho(i)}, x_{\lambda(i)})}. \end{aligned}$$

We now have a ratio of two intractable integrals, over relatively low-dimensional spaces, but with a priori intractable integrands. The proposal is then to use the locality of the underlying random field to synthesise an approximate distribution which can then be marginalised.

In more concrete terms, assume that the joint law of the random field can be written in the form

$$\mathbf{P}(dx) = \mathbf{P}_0(dx) \cdot \frac{\exp(-\mathcal{H}(x))}{\mathcal{Z}}$$

$$\mathcal{H}(x) = \sum_{a \in \mathcal{F}} \mathcal{H}_a(x_{\partial a})$$

for some product measure $\mathbf{P}_0$ and some ‘factor decomposition’ $\mathcal{F}$. For each vertex i , construct then

$$\mathcal{H}_i(x) = \sum_{a \in \mathcal{F}_i} \mathcal{H}_a(x_{\partial a})$$

$$\mathcal{F}_i = \{a \in \mathcal{F} : \partial a \subseteq \nu(i)\}$$

$$\nu(i) = \lambda(i) \sqcup \{i\} \sqcup \rho(i),$$

where $\nu(i) \subseteq \mathcal{V}$ now denotes a ‘neighbourhood’ of the vertex i , looking both forwards and backwards, and $\mathcal{H}_i(x)$ entails the terms in the energy function $\mathcal{H}$ which involve only the variables in $\nu(i)$. Note that since the decomposition of $\mathcal{H}$ into the sum of the $\mathcal{H}_a$ is non-canonical (i.e. it is really a choice of the statistician), there is a degree of arbitrariness in this construction. Nevertheless, natural choices often present themselves in applications.

Having made these choices, one can then approximate

$$\mathbf{P}(dx_{\nu(i)}) \approx \mathbf{P}^{(i)}(dx_{\nu(i)}) := \mathbf{P}_0(dx_{\nu(i)}) \cdot \frac{\exp(-\mathcal{H}_i(x))}{\mathcal{Z}_i}$$

That is, assuming that the probability measure of interest is a Gibbs measure, we decompose the associated Hamiltonian into its local contributions, identify the contributions which depend only on the most pertinent other vertices, and neglect all other terms. One can then approximate the constituent distributions of the Vecchia approximation by arguing that

$$\mathbf{P}(dx_i | x_{\lambda(i)}) = \frac{\int_{\mathcal{X}_{\rho(i)}} \mathbf{P}(dx_i, d\bar{x}_{\rho(i)}, x_{\lambda(i)})}{\int_{\mathcal{X}_{i \sqcup \rho(i)}} \mathbf{P}(d\bar{x}_i, d\bar{x}_{\rho(i)}, x_{\lambda(i)})} \approx \frac{\int_{\mathcal{X}_{\rho(i)}} \mathbf{P}^{(i)}(dx_i, d\bar{x}_{\rho(i)}, x_{\lambda(i)})}{\int_{\mathcal{X}_{i \sqcup \rho(i)}} \mathbf{P}^{(i)}(d\bar{x}_i, d\bar{x}_{\rho(i)}, x_{\lambda(i)})} =: \mathbf{P}^{\text{OCA}}(dx_i | x_{\lambda(i)}),$$

where these integrals can be computed under the assumption that $\mathbf{P}^{(i)}$ is sufficiently simple. Thus far, the method has apparently only been applied when $\mathbf{P}^{(i)}$ is a probability distribution over a discrete state space of modest size. It remains to be seen whether other examples will be forthcoming.

Continuous OCA by Laplace Approximation

Suppose that each x_i is real-valued (the $\mathbf{R}^d$ -valued case is analogous, but more notationally intensive), and the energy function is composed of sufficiently smooth functions. In this context, one can consider the use of mode-centered Gaussian approximations to approximate intractable integrals, i.e. given $(x_i, x_{\lambda(i)})$,

1. Write $\mathbf{m}_{\rho|i,\lambda}(x_{i \sqcup \lambda(i)})$ for the minimiser of $\bar{x}_{\rho(i)} \mapsto \mathcal{H}_i(x_{i \sqcup \lambda(i)}, \bar{x}_{\rho(i)})$, develop the Taylor expansion

$$\mathcal{H}_i(x_{i \sqcup \lambda(i)}, \bar{x}_{\rho(i)}) \simeq \mathcal{H}_i(x_{i \sqcup \lambda(i)}, \mathbf{m}_{\rho|i,\lambda}) + \frac{1}{2} \cdot \mathcal{H}_i''(x_{i \sqcup \lambda(i)}, \mathbf{m}_{\rho|i,\lambda}) \cdot [\bar{x}_{\rho(i)} - \mathbf{m}_{\rho|i,\lambda}]^2,$$

and hence infer the Laplace approximation

$$\int_{\mathcal{X}_{\rho(i)}} \mathbf{P}^{(i)}(dx_i, d\bar{x}_{\rho(i)}, x_{\lambda(i)}) \approx \sqrt{\frac{2\pi}{\mathcal{H}_i''(x_{i \sqcup \lambda(i)}, \mathbf{m}_{\rho|i,\lambda})}} \cdot \exp(-\mathcal{H}_i(x_{i \sqcup \lambda(i)}, \mathbf{m}_{\rho|i,\lambda}))$$

$$=: \mathcal{Z}_{i,\lambda}^{\text{LA}}(x_{i \sqcup \lambda(i)}) \cdot \exp(-\mathcal{H}_{i \sqcup \lambda}(x_{i \sqcup \lambda(i)})).$$

2. Write $\mathbf{m}_{\rho,i|\lambda}(x_{\lambda(i)})$ for the minimiser of $\bar{x}_{\rho(i) \sqcup i} \mapsto \mathcal{H}_i(x_{\lambda(i)}, \bar{x}_{\rho(i) \sqcup i})$, develop the Taylor expansion

$$\mathcal{H}_i(x_{\lambda(i)}, \bar{x}_{\rho(i) \sqcup i}) \simeq \mathcal{H}_i(x_{\lambda(i)}, \mathbf{m}_{\rho,i|\lambda}) + \frac{1}{2} \cdot \mathcal{H}_i''(x_{\lambda(i)}, \mathbf{m}_{\rho,i|\lambda}) \cdot [\bar{x}_{\rho(i) \sqcup i} - \mathbf{m}_{\rho,i|\lambda}]^2,$$

and hence infer the Laplace approximation

$$\int_{\mathcal{X}_{i \sqcup \rho(i)}} \mathbf{P}^{(i)}(d\bar{x}_i, d\bar{x}_{\rho(i)}, x_{\lambda(i)}) \approx \sqrt{\frac{2\pi}{\mathcal{H}_i''(x_{\lambda(i)}, \mathbf{m}_{\rho,i|\lambda})}} \cdot \exp(-\mathcal{H}_i(x_{\lambda(i)}, \mathbf{m}_{\rho,i|\lambda}))$$

$$=: \mathcal{Z}_{\lambda \setminus i}^{\text{LA}}(x_{\lambda(i)}) \cdot \exp(-\mathcal{H}_{\lambda \setminus i}(x_{\lambda(i)})).$$

3. Assemble the approximation

$$\begin{aligned} \mathbf{P}(\mathrm{d}x_i \mid x_{\lambda(i)}) &\approx \frac{\mathcal{Z}_{i,\lambda}^{\text{LA}}(x_{i \sqcup \lambda(i)}) \cdot \exp(-\mathcal{H}_{i \sqcup \lambda}(x_{i \sqcup \lambda(i)}))}{\mathcal{Z}_{\lambda \setminus i}^{\text{LA}}(x_{\lambda(i)}) \cdot \exp(-\mathcal{H}_{\lambda \setminus i}(x_{\lambda(i)}))} \\ &= \frac{\mathcal{Z}_{i,\lambda}^{\text{LA}}(x_{i \sqcup \lambda(i)})}{\mathcal{Z}_{\lambda \setminus i}^{\text{LA}}(x_{\lambda(i)})} \cdot \exp(-[\mathcal{H}_{i \sqcup \lambda}(x_{i \sqcup \lambda(i)}) - \mathcal{H}_{\lambda \setminus i}(x_{\lambda(i)})]) =: \mathbf{P}^{\text{OCLA}}(\mathrm{d}x_i \mid x_{\lambda(i)}). \end{aligned}$$

In this instance, computing the ‘Ordered Conditional Laplace Approximation’ may be computationally feasible. Computing derivatives of $\log p^{\text{OCLA}}$ becomes somewhat more cumbersome, but should be amenable to (careful) treatment by methods of automatic and implicit differentiation.

Some More Error Analysis

We can study the error incurred by this approximation. Define $\mathcal{H}_i^{\mathfrak{C}}(x) = \mathcal{H}(x) - \mathcal{H}_i(x)$ to be the ‘complementary’ energy which is neglected by our approximation. Compute then that the marginal distribution of $\{x_j : j \in \nu(i)\}$ is expressible as

$$\begin{aligned} \mathbf{P}(\mathrm{d}x_{\nu(i)}) &= \int \mathbf{P}_0(\mathrm{d}x_{\nu(i)}) \cdot \mathbf{P}_0(\mathrm{d}\bar{x}_{\nu(i)\mathfrak{C}} \mid x_{\nu(i)}) \cdot \frac{\exp(-\mathcal{H}_i(x_{\nu(i)}) - \mathcal{H}_i^{\mathfrak{C}}(x_{\nu(i)}, \bar{x}_{\nu(i)\mathfrak{C}}))}{\mathcal{Z}} \\ &= \mathbf{P}_0(\mathrm{d}x_{\nu(i)}) \cdot \frac{\exp(-\mathcal{H}_i(x_{\nu(i)}))}{\mathcal{Z}_i} \cdot \int \mathbf{P}_0(\mathrm{d}\bar{x}_{\nu(i)\mathfrak{C}} \mid x_{\nu(i)}) \cdot \frac{\exp(-\mathcal{H}_i^{\mathfrak{C}}(x_{\nu(i)}, \bar{x}_{\nu(i)\mathfrak{C}}))}{\mathcal{Z}/\mathcal{Z}_i} \\ &=: \mathbf{P}_0(\mathrm{d}x_{\nu(i)}) \cdot \frac{\exp(-\mathcal{H}_i(x_{\nu(i)}) - \delta\mathcal{H}_i(x_{\nu(i)}))}{\mathcal{Z}_i}, \end{aligned}$$

where

$$\delta\mathcal{H}_i(x_{\nu(i)}) = \log\left(\frac{\mathcal{Z}}{\mathcal{Z}_i}\right) - \log\left(\int \mathbf{P}_0(\mathrm{d}\bar{x}_{\nu(i)\mathfrak{C}} \mid x_{\nu(i)}) \cdot \exp(-\mathcal{H}_i^{\mathfrak{C}}(x_{\nu(i)}, \bar{x}_{\nu(i)\mathfrak{C}}))\right).$$

We hence see that the implied approximation $\mathbf{P}^{(i)}$ can be expected to be accurate when the function $\delta\mathcal{H}_i$ is approximately constant. Naturally, this is implied by $\bar{x}_{\nu(i)\mathfrak{C}} \mapsto \mathcal{H}_i^{\mathfrak{C}}(x_{\nu(i)}, \bar{x}_{\nu(i)\mathfrak{C}})$ being approximately constant for each $x_{\nu(i)}$.

Perspectives on the Ordered Conditional Approximation

Through this lens, we can view the OCA as making a sequence of approximations in service of tractability:

1. (Sparsify Dependence on the Past) Approximate

$$\mathbf{P}(\mathrm{d}x_i \mid x_{\prec i}) \approx \mathbf{P}(\mathrm{d}x_i \mid x_{\lambda(i)}).$$

2. (Neglect Superficially Irrelevant Factors) Approximate

$$\mathbf{P}(\mathrm{d}x_{\nu(i)}) \approx \mathbf{P}_0(\mathrm{d}x_{\nu(i)}) \cdot \frac{\exp(-\mathcal{H}_i(x_{\nu(i)}))}{\mathcal{Z}_i} =: \mathbf{P}^{(i)}(\mathrm{d}x_{\nu(i)}).$$

3. (Account for Influence of the Future) Infer the approximation

$$\mathbf{P}(\mathrm{d}x_i \mid x_{\lambda(i)}) \approx \frac{\int \mathbf{P}^{(i)}(\mathrm{d}x_i, \mathrm{d}\bar{x}_{\rho(i)}, x_{\lambda(i)})}{\int \mathbf{P}^{(i)}(\mathrm{d}\bar{x}_i, \mathrm{d}\bar{x}_{\rho(i)}, x_{\lambda(i)})} =: \mathbf{P}^{(i)}(\mathrm{d}x_i \mid x_{\lambda(i)}).$$

4. (Aggregate Local Models) Infer the approximation

$$\mathbf{P}(\mathrm{d}x) \approx \prod_{i \in \mathcal{V}} \mathbf{P}^{(i)}(\mathrm{d}x_i \mid x_{\lambda(i)}) =: \mathbf{P}^{\text{OCA}}(\mathrm{d}x).$$

Conclusion and Comments

A number of additional points are worthy of further discussion:

1. A nice feature of the Vecchia and Ordered Conditional Approximations is that they each define genuine normalised probability distributions by virtue of their construction in terms of locally-normalised directed graphical models. The Ordered Conditional Laplace Approximation is not guaranteed to enjoy this property without e.g. an additional step of univariate numerical integration.
2. The cursory analysis presented in this note is rather too worst-case to be of much theoretical interest. It would be useful to identify some more practical conditions of the form ‘if the density $[\cdot \cdots]$ is approximately constant in some average sense, then the approximation performs well’.
3. We do not really discuss how to come up with good λ, ρ in practice. Some knowledge about this issue has accrued over time, which can be rather important in practice. As an example, for spatial models, a number of nearest-neighbour-type heuristics have seen widespread adoption.